

Supplementary Material: Environment Sensitivity and Reproducibility of Post-BA SfM Benchmarks

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A. What this document supports

This supplementary substantiates four specific statements in the main paper; a reviewer following any of its pointers can go directly to the listed section.

- Protocol, “Track provenance”: “ ~ 0.209 is the reproducible level [for the released RESfM checkpoint] ... unchanged (0.217) under the authors’ exact pinned software stack” (Secs. B–D).
- Protocol: “a stack-replication study (supplementary) shows the no-TTT arms are insensitive to it, while test-time fine-tuned means inherit solver-level nondeterminism” (Sec. E).
- Table 2, † footnote: “TTT is run-to-run nondeterministic” (Sec. E).
- Reading any post-BA mean in this literature: differences $\lesssim 0.05$ are within environment noise (Sec. F).

In addition, Sec. G tabulates the per-scene translation, rotation, and registered-camera (N_r) results behind the main paper’s dataset-level numbers, with dataset-level rotation and registration summaries in Table 11; Sec. H gives the complete training and evaluation configuration, and Sec. I provides the per-scene label-degradation breakdown behind the label-ceiling analysis.

B. The reproduction question

Evaluating the RESfM authors’ released checkpoint in our harness on the 36 MegaDepth test scenes yields a post-BA mean translation error of 0.209 vs. the published 0.169 (rotation 1.89° vs. 1.29°). Since our controlled comparisons hinge on baseline fidelity, we traced this discrepancy to its source. The finding is of independent interest: post-BA means on internet-photo benchmarks carry a practical environment-noise floor of $\mathcal{O}(0.05)$, produced by chaotic basin selection in robust bundle adjustment.

Table 8. Released RESfM checkpoint, MegaDepth 36-scene post-BA means, under three conditions, vs. the published number. N_r is the mean number of registered cameras per scene (mean $N_c=364.6$ ground-truth cameras); the published row averages the paper’s per-scene Table 1 values.

Condition	Trans	Rot ($^\circ$)	N_r
Published (paper Table 1)	0.169	1.29	298.9
Our env (pip/OpenBLAS), 5-seed mean	0.203	1.82	303.9
Our env, seed 20	0.209	1.89	303.6
Authors’ exact stack (conda/MKL)	0.217	1.92	303.7

C. Eliminating the controllable factors

Data. Our MegaDepth npz files are byte-identical to the authors’ released archive, including the Group-2 300-camera subsamples.

Randomness. Five evaluation seeds vary the pre-BA network output by ± 0.004 and the post-BA mean by ± 0.01 ; negligible.

Network forward pass. The pre-BA output (translation/rotation $\sim 1.73/11.9^\circ$) is essentially seed- and environment-stable; the divergence therefore enters at the robust-BA stage. (Notably, the paper does not report pre-BA numbers; all its tables are post-robust-BA.)

Software stack. We rebuilt the authors’ exact pinned environment from their released environment.yaml: Python 3.8.20, MKL-backed numpy 1.24.3 (mkl=2023.1.0), pycolmap 3.10.0, pyceres 2.3, lightning 2.3.3. (The sole deviation is the CUDA build of PyTorch, forced by GPU-driver compatibility; the BA stage is CPU-only.) Re-evaluating all 36 scenes in this matched environment gives translation/rotation errors of $0.217/1.92^\circ$, statistically indistinguishable from our environment’s $0.209/1.89^\circ$, and not the published $0.169/1.29^\circ$. Registration is equally stable: all three of our conditions register 303.6–303.9 cameras per scene on average (Table 8), slightly more than the published

062	298.9, so the higher mean error is not an artifact of		
063	evaluating fewer cameras.		
064	D. Mechanism: chaotic basin selection in ro-		
065	bust BA		
066	Initialization perturbation is at the 10^{-10} level. We		
067	isolated the BLAS layer by running the identical DLT		
068	triangulation (the BA initialization) on identical inputs		
069	under OpenBLAS and MKL, single-threaded: over		
070	11,131 triangulated points the maximum relative dif-		
071	ference is 4.8×10^{-10} (mean 5×10^{-14} ; NaN patterns		
072	identical).		
073	Yet outcomes move by $\mathcal{O}(0.1)$. This sub-nanoscale per-		
074	turbation changes the per-scene post-BA translation by		
075	up to 0.167, with 16/36 scenes moving by more than		
076	0.01. Robust BA on ill-conditioned scenes amplifies		
077	last-bit differences into different convergence basins.		
078	The published gap has the same signature. Compar-		
079	ing our released-checkpoint evaluation to the paper's		
080	Table 1 per scene: 20/36 scenes agree within 0.05 (8		
081	essentially exactly), 17 are better in our runs, and the		
082	entire mean gap is carried by four basin flips (scenes		
083	0046, 0148, 0012, 5016; e.g. 0046: $0.043 \rightarrow 0.892$).		
084	Bidirectional, scene-concentrated flips are the signa-		
085	ture of basin selection, not of a weaker checkpoint,		
086	which would shift errors in one direction rather than		
087	improve 17 of 36 scenes.		
088	E. Stack replication and TTT run-to-run vari-		
089	ance		
090	Beyond the released checkpoint, we re-evaluated both		
091	from-scratch main-comparison checkpoints under the		
092	authors' pinned stack (MKL numpy, pycolmap 3.10.0,		
093	pyceres 2.3), changing nothing else, and additionally		
094	repeated the TTT evaluation within our own stack.		
095	The supervised baseline is essentially unchanged across		
096	stacks: MegaDepth $0.344 \rightarrow 0.350$ mean, $0.070 \rightarrow$		
097	0.088 median, only 7/36 scenes moving more than 0.05		
098	(max ± 0.37 , bidirectional). This matches the released-		
099	checkpoint result above. The SS-RESfM weight+TTT		
100	arm, by contrast, varies regardless of stack: three eval-		
101	uations of the same checkpoint (ours, a same-stack re-		
102	peat, the authors' stack) give MegaDepth means 0.316,		
103	0.401, 0.453 (~ 18 – 20 /36 scenes shifting per repeat, up		
104	to ± 2.9 , largely the same scenes, medians 0.12–0.19).		
105	The attribution is therefore not the software stack but		
106	test-time fine-tuning itself: it runs an optimization at		
107	evaluation time, so the chaotic basin selection docu-		
108	mented above re-enters the result on every run. Con-		
109	sequently the paper reports all results from the sin-		
110	gle declared pipeline, treats TTT means as carrying		
111	$\sim \pm 0.07$ run-to-run variance, and rests its environment-		
	and run-independent claims on the deterministic no-		
	TTT cells (which reproduce exactly on repeat).		
	F. Implications for reading the paper's num-		
	bers		
	(1) The published number and ours are draws from		
	a distribution over environments; no software match-		
	ing can force a particular draw. (2) Differences		
	of $\lesssim 0.05$ in post-BA translation means on these		
	benchmarks are within environment noise and should		
	not be read as method differences, by any paper.		
	(3) Consequently, meaningful comparisons must be		
	environment-controlled: every number in the main pa-		
	per (all arms, RESfM-from-scratch, and the released		
	checkpoint) is produced in a single environment, and		
	the out-of-distribution margins ($2\times$ and larger) sit far		
	above this noise floor.		
	G. Per-scene results		
	Tables 9 and 10 report the per-scene post-BA trans-		
	lation and rotation errors and registered-camera frac-		
	tions underlying the main paper's dataset-level num-		
	bers (seed 20, the declared pipeline): the ten		
	1DSfM scenes and the 36 MegaDepth test scenes, for		
	the three self-supervised arms (madweight, weight,		
	weight+TTT) and the supervised RESfM baseline		
	trained from scratch under the identical recipe. The		
	per-scene view makes two aggregate statements con-		
	crete: on 1DSfM no single arm dominates, the best		
	mechanism flips scene by scene (with <u>Ellis_Island</u> and		
	<u>Tower_of_London</u> hard for every arm), consistent		
	with the per-scene complementarity analysis in the		
	main paper; on MegaDepth the four arms are close		
	on most scenes and the dataset-level gap is dominated		
	by a handful of high-error scenes. Per scene, the		
	catastrophic scenes (<u>Ellis_Island</u> , <u>Tower_of_London</u>)		
	fail in both metrics, though rotation does not always		
	track translation (weight's <u>Vienna_Cathedral</u> failure		
	is translation-specific: $22.71/0.46$), and N_r stays in		
	the 58–92% range on 1DSfM with no arm systemati-		
	cally dropping scenes. Table 11 completes the dataset-		
	level record with rotation and registered-camera counts		
	alongside translation. Rotation broadly agrees with		
	translation at the dataset level: the supervised Blend-		
	edMVS failure is even larger in rotation (36°) and		
	the high-contamination ordering is preserved, though		
	the best arm can differ per metric (on MegaDepth,		
	weight+TTT leads translation while RESfM leads ro-		
	tation). Registration is contamination-dependent: the		
	supervised baseline registers the most cameras in distri-		
	bution (85% vs. 78–80% on MegaDepth) and on clean		
	Strecha, while under high contamination the label-free		

Table 9. Per-scene 1DSfM results. Post-BA translation and rotation error per scene (camera means; rotation in degrees), registered-camera fraction N_r , and the scene’s ground-truth camera count N_c , seed 20. Best translation per scene in bold.

Scene	madweight		weight		weight+TTT		RESfM		N_c
	Trans/Rot	N_r	Trans/Rot	N_r	Trans/Rot	N_r	Trans/Rot	N_r	
Alamo	1.77/2.45	81%	2.18/2.74	88%	1.67 /2.49	86%	1.73/2.42	87%	577
Ellis_Island	17.59/17.16	79%	15.90 /17.37	75%	17.11/17.74	78%	16.02/18.15	85%	227
Madrid_Metropolis	9.63/5.72	73%	4.37/2.84	88%	3.52/2.35	87%	3.38 /1.82	84%	341
Montreal_Notre_Dame	0.94/0.59	69%	11.35/0.55	72%	1.24/0.81	71%	0.93 /0.59	89%	450
Notre_Dame	0.68/3.33	92%	0.34/1.00	89%	0.33 /0.83	88%	0.49/1.08	89%	553
NYC_Library	2.94/2.32	85%	1.19 /1.16	79%	3.39/10.30	84%	6.28/12.99	86%	332
Piazza_del_Popolo	0.94 /0.43	64%	10.66/19.54	78%	17.05/20.46	75%	2.49/3.15	75%	338
Tower_of_London	76.34/35.61	67%	28.29 /23.30	71%	72.61/17.91	67%	67.95/49.66	63%	472
Vienna_Cathedral	7.30/7.42	64%	22.71/0.46	58%	33.57/26.80	78%	2.29 /1.09	59%	836
Yorkminster	5.79 /8.05	68%	30.87/14.95	85%	10.79/12.20	87%	8.77/6.88	71%	437

arms retain more cameras than supervision (38–43% vs. 37% on 1DSfM-hard) at lower error (madweight).

H. Implementation details

Table 12 lists the full training and evaluation configuration. Both methods share every setting except the base learning rate (10^{-4} self-supervised, 10^{-3} supervised, each selected on its own training objective) and the supervision source (COLMAP-derived labels vs. the adaptive percentile pseudo-labels).

Table 10. Per-scene MegaDepth results. Post-BA translation and rotation error per scene (camera means; rotation in degrees), registered-camera fraction N_r , and N_c , seed 20. Best translation per scene in bold. weight+TTT values are single draws of a run-to-run nondeterministic evaluation (Sec. E).

Scene	madweight		weight		weight+TTT		RESfM		N_c
	Trans/Rot	N_r	Trans/Rot	N_r	Trans/Rot	N_r	Trans/Rot	N_r	
0238	1.41/19.02	84%	0.62/8.23	26%	0.63/8.37	27%	0.12 /1.40	68%	522
0060	0.03/0.29	89%	0.02/0.20	90%	0.37/4.13	86%	0.01 /0.04	94%	528
0197	0.89/7.22	54%	0.38/3.54	39%	1.11/6.94	49%	0.11 /0.97	85%	870
0094	0.45/2.72	65%	0.16 /1.44	66%	0.17/1.34	57%	0.88/5.33	75%	763
0265	3.26/17.62	80%	2.34/5.78	84%	0.19 /0.60	96%	3.09/16.27	74%	571
0083	0.23/4.34	54%	1.56/17.70	80%	0.20/3.20	50%	0.10 /1.40	92%	635
0076	0.38/1.40	76%	0.23/0.61	93%	0.22/0.74	81%	0.02 /0.25	89%	558
0185	0.01/0.21	94%	0.04/0.24	91%	0.09/0.88	87%	0.01 /0.07	96%	368
0048	0.14/1.93	93%	0.02 /0.35	91%	0.03/0.43	88%	0.11/1.85	96%	512
0024	0.18/1.27	84%	0.17 /1.13	82%	1.13/4.78	81%	1.09/3.67	94%	356
0223	0.24/2.66	96%	0.23/1.69	93%	0.30/3.06	93%	0.08 /0.45	95%	214
5016	0.04/0.27	82%	0.02/0.08	89%	0.02/0.07	100%	0.01 /0.08	100%	28
0046	0.22 /2.88	96%	0.30/4.11	96%	0.25/3.88	95%	0.93/14.64	93%	440
0099	0.15/0.81	75%	3.52/2.77	72%	0.13 /1.58	71%	0.39/2.18	84%	299
1001	1.22/1.58	98%	0.73/1.49	98%	0.49 /1.22	98%	0.50/1.23	96%	285
0231	0.15/1.63	93%	0.07/0.64	87%	0.24/1.49	89%	0.02 /0.17	90%	296
0411	0.03/0.22	94%	0.32/2.99	82%	0.46/4.12	84%	0.02 /0.11	92%	299
0377	0.05/0.62	68%	0.26/2.76	75%	0.03 /0.56	82%	0.06/0.82	76%	295
0102	0.02/0.15	97%	0.01 /0.07	97%	0.01/0.08	97%	0.02/0.10	96%	299
0147	2.99/31.69	88%	1.21/15.28	79%	1.20 /15.27	76%	3.22/4.61	83%	298
0148	0.33/2.60	77%	0.11/2.11	70%	0.04/0.62	68%	0.04 /0.94	70%	287
0446	0.03 /0.40	98%	0.19/1.20	96%	0.19/1.04	98%	0.05/0.71	98%	298
0022	0.02/0.14	93%	0.03/0.22	92%	0.04/0.33	90%	0.02 /0.11	93%	297
0327	0.09/1.97	95%	0.02/0.22	96%	0.02/0.21	95%	0.01 /0.09	93%	298
0015	0.30/2.64	69%	0.56/3.87	69%	0.65/3.90	74%	0.30 /1.31	82%	284
0455	0.01 /0.20	95%	0.01/0.21	96%	1.60/0.46	97%	0.09/0.85	98%	298
0496	0.04/0.27	94%	0.04 /0.50	96%	0.09/1.28	97%	0.07/0.46	95%	297
1589	0.06/0.96	89%	0.05/0.28	88%	0.30/1.91	99%	0.01 /0.06	97%	299
0012	0.02 /0.23	94%	0.70/9.51	83%	0.22/2.76	88%	0.34/5.53	96%	299
0104	0.03/0.32	67%	0.18/2.28	68%	0.07/1.44	68%	0.02 /0.30	70%	284
0019	0.17/3.22	78%	0.28/6.55	85%	0.13/3.24	88%	0.01 /0.05	84%	299
0063	0.08/0.63	88%	0.07/0.72	91%	0.06 /0.65	91%	0.07/0.44	91%	293
0130	1.60/10.17	88%	0.16/1.87	65%	0.01 /0.33	65%	0.07/1.03	68%	285
0080	0.05 /1.10	46%	0.20/2.48	52%	0.07/0.70	52%	0.10/0.66	50%	284
0240	0.25 /2.23	92%	0.34/4.64	92%	0.46/5.32	91%	0.35/3.02	95%	298
0007	0.01 /0.07	37%	0.09/1.86	93%	0.14/3.28	90%	0.04/0.99	36%	290

Table 11. Dataset-level translation, rotation, and registered cameras (post-BA, seed 20) for the four main-comparison arms. Per-scene values are camera means; N_r/N_c is the fraction of ground-truth cameras surviving robust BA and the 5-pixel reprojection quality mask, summed over the dataset’s scenes.

	madweight	weight	weight+TTT	RESfM
Translation error (mean / median over scenes)				
MegaDepth	0.422/0.145	0.423/0.182	0.316/0.186	0.344/0.07
1DSfM	12.4/4.37	12.8/11.0	16.1/7.16	11.0/2.93
1DSfM-hard	12.2/8.83	21.9/13.1	26.4/17.7	16.2/13.1
Strecha	2.5/0.97	2/0.026	1.98/0.005	0.006/0.005
BlendedMVS	0.146/0.004	0.1/0.036	0.136/0.011	0.379/0.348
Olsson	2.57/1.15	2.65/1.22	7.74/0.18	8.47/0.17
Rotation error ($^\circ$, mean / median over scenes)				
MegaDepth	3.49/1.33	3.04/1.77	2.51/1.39	2.01/0.84
1DSfM	8.31/4.53	8.39/2.79	11.19/11.25	9.79/2.79
1DSfM-hard	14.40/7.39	25.58/17.70	31.97/12.83	22.70/15.86
Strecha	14.62/8.43	7.48/0.17	7.41/0.02	0.02/0.01
BlendedMVS	9.60/0.38	6.65/3.06	7.91/0.55	36.10/29.55
Olsson	13.57/4.37	13.23/5.41	10.49/2.27	12.34/3.35
Registered cameras N_r/N_c				
MegaDepth	80%	79%	78%	85%
1DSfM	74%	77%	80%	77%
1DSfM-hard	38%	39%	43%	37%
Strecha	91%	96%	98%	100%
BlendedMVS	95%	88%	95%	90%
Olsson	83%	90%	90%	88%

Table 12. Training and evaluation configuration. All arms in the main paper use these settings; the only per-method difference is the base learning rate.

Setting	Value
Training (both methods)	
Epochs	20,000 (one epoch = one pass over all scenes)
Batch	4 scenes
LR schedule	milestones 10k, 11k, . . . , 20k, $\gamma = 0.9$
Base LR	10^{-4} (SS-RESfM) / 10^{-3} (RESfM)
Loss weights	$\alpha_{\text{reproj}} = 1$, $\beta_{\text{cls}} = 1$
Infinity-point margin	10^{-4}
Checkpoint selection	best validation error, evaluated every 500 epochs
Seed	20 (21–22 / 21–24 for the seed studies)
Adaptive pseudo-labels (SS-RESfM only)	
Inlier / outlier percentile	20 / 80 (of per-observation reproj. error)
Threshold separation	scale-aware floor, $0.25\times$ median error
Warm-up	10 epochs before pseudo-labeling starts
Evaluation	
Removal threshold τ	0.6 (0.8 for low-contamination single-camera sets, following RESfM)
Test-time fine-tuning	1,001 iterations, LR 5×10^{-3} , evaluated every 250 iterations
Bundle adjustment	pyceres/pycolmap robust BA, as RESfM

I. Per-scene label degradation

Table 13 gives the per-scene data behind the main paper’s label-ceiling analysis: for each of the 18 scenes we measure the contamination level, the fraction of track observations that receive a COLMAP-derived label at all (coverage), and the precision and recall of those labels against ground truth. The aggregate trend in the main paper is visible scene by scene: below $\sim 10\%$ contamination coverage is 0.92–1.00 and recall is 0.66–0.97 (mostly above 0.8), while above $\sim 15\%$ contamination recall never exceeds 0.50 on any scene (0.25–0.50) and coverage falls as low as 0.37: the supervision signal degrades exactly where outlier handling matters most.

Table 13. Per-scene label-degradation results. For each scene: contamination (fraction of track observations that are GT outliers), COLMAP-label coverage (fraction of observations receiving a label), and precision/recall of the COLMAP-derived outlier labels against GT, sorted by contamination. Recall collapses above $\sim 15\%$ contamination.

Dataset	Scene	Contam. (%)	Coverage	Prec.	Recall
BlendedMVS	58c4bb4f4a...	0.5	1.00	0.86	0.94
Strecha	Herz-Jesu-P8	0.7	1.00	0.51	0.66
Strecha	fountain-P11	0.7	1.00	0.69	0.76
Strecha	Herz-Jesu-P25	2.6	0.99	0.72	0.82
Strecha	entry-P10	2.7	0.99	0.88	0.88
BlendedMVS	5be883a4f9...	4.0	0.98	0.90	0.84
BlendedMVS	5b6e716d67...	6.1	0.92	0.89	0.92
BlendedMVS	5bfd0f32ec...	7.1	0.97	0.97	0.97
BlendedMVS	5b950c7160...	9.2	0.96	0.92	0.94
MegaDepth	0007	11.2	0.73	0.84	0.47
1DSfM	Alamo	26.6	0.76	0.81	0.25
1DSfM	Madrid_Metropolis	28.8	0.67	0.85	0.28
MegaDepth	0041	31.4	0.49	0.80	0.42
MegaDepth	0060	39.0	0.42	0.53	0.45
1DSfM	Piazza_del_Popolo	41.9	0.76	0.90	0.30
MegaDepth	0047	54.9	0.37	0.83	0.50
1DSfM	Ellis_Island	57.9	0.79	0.89	0.37
1DSfM-hard	Gendarmenmarkt	59.8	0.50	0.89	0.29